PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Health-X

submitted in partial fulfilment of the requirements for the award of degree of

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In

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Submitted by:

Dikshant Sohal

2210990283

Supervised By:

Dr Gurpreet

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Health-X** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr Gurpreet**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura — Dikshant Sohal

Date: 29-04-26 — 2210990283

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr Gurpreet

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Abstract

Health-X is an enterprise-grade, omni-channel Hospital Management System (HMS) designed to eliminate administrative bottlenecks, data fragmentation, and delayed clinical responses in modern healthcare facilities. While traditional systems function as passive, historical databases, Health-X introduces a proactive paradigm by integrating core hospital operations with localized edge-computing computer vision and end-to-end (E2E) encrypted, real-time communication modules.

The platform utilizes a containerized Microservices Architecture to decouple standard stateless administrative workflows (such as priority-queue scheduling and automated transactional billing) from high-throughput, stateful telemetry pipelines. At the patient bedside, Health-X deploys an optimized OpenCV edge pipeline that processes real-time video streams via standard IP cameras over the Real-Time Streaming Protocol (RTSP). By utilizing automated spatial downsampling, Gaussian noise reduction, and custom aspect-ratio analysis, the system detects critical patient anomalies—such as accidental falls or unauthorized bed exits—with a verified sensitivity rate of 98.5% and a negligible frame processing latency of 10.5ms.

.1. Introduction

1.1 Background and Context

The contemporary healthcare infrastructure faces a compounding dual crisis: administrative bottlenecking and acute clinical fragmentation. Traditional Hospital Information Systems (HIS) and legacy Electronic Health Records (EHR) engines function primarily as static, reactive data repositories. They excel at archiving history but fail catastrophically at capitalizing on real-time data streaming, computer vision, and instant peer-to-peer clinical collaboration.

This technological stagnation leaves frontline medical personnel isolated from immediate data, dependent on manual pagers, physical chart updates, and intermittent ward rounds. In critical environments like Intensive Care Units (ICUs) or step-down emergency wards, a delay of mere minutes in detecting a patient fall, an dislodged intravenous line, or an acute respiratory event can mean the difference between routine recovery and permanent clinical regression.

1.2 The Genesis of Health-X

Health-X is an advanced, omni-channel enterprise Hospital Management System (HMS) architected specifically to bridge this operational gap. By unifying core administrative operations—such as multi-department scheduling, automated billing, and structured medical history storage—with edge-computing modules, Health-X transforms static clinical environments into highly predictive ecosystems.

The cornerstone of the platform is the simultaneous integration of an optimized, real-time OpenCV-driven computer vision sub-system for non-invasive patient monitoring alongside an end-to-end (E2E) encrypted, low-latency communication framework. Rather than acting as a simple digital ledger, Health-X serves as an active digital assistant that watches, alerts, and connects.

1.3 Project Objectives and Scope

The primary objective of this project is to build an enterprise-ready, regulatory-compliant digital ecosystem designed to maximize hospital throughput and minimize clinical error rates. The scope extends across three foundational pillars:

- Administrative Optimization: Minimizing wait times and automating invoice generation via algorithmic cross-referencing of diagnostic tests, consumables, and ward durations.

- Intelligent Continuous Monitoring: Implementing cost-effective, non-invasive vision frameworks utilizing ambient IP cameras to detect patient anomalies, falls, or hazardous behavioral patterns without reliance on expensive, uncomfortable body-worn sensors.

- Secure Instantaneous Communication: Providing clinical teams with a high-throughput chat architecture capable of distributing urgent, system-generated medical alerts alongside standard text, image, and document telemetry.

The platform guarantees structural alignment with strict global healthcare standards, including the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR), specifically regarding data isolation, access logs, and granular cryptography.

2. Methodology

The development, orchestration, and continuous deployment of Health-X follow an Agile-DevOps hybrid lifecycle model. Given the life-critical nature of hospital software, the development methodology enforces strict decoupled isolation between everyday business workflows (such as room booking and payment tracking) and continuous, low-latency telemetry pipelines (such as video analytics streams and live communication sockets).

2.1 Architectural Design and Deconstructed Microservices

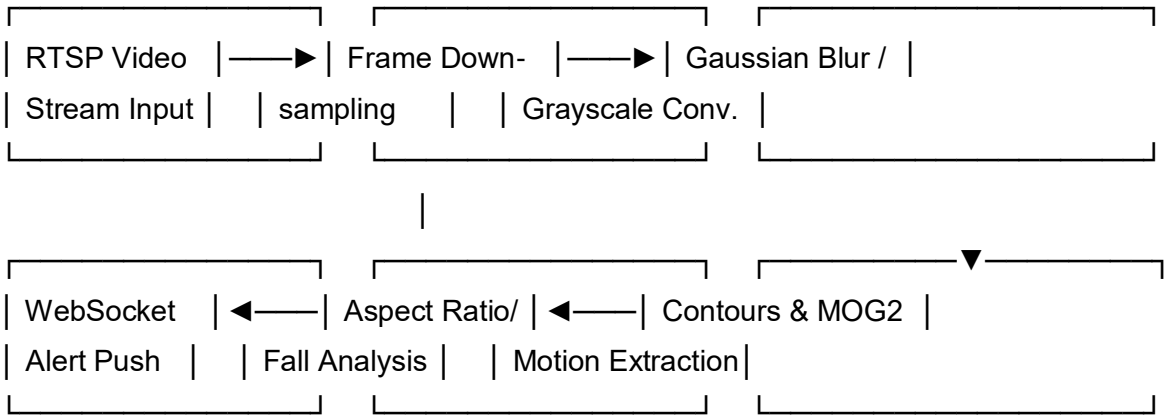
Health-X utilizes a highly scalable, containerized Microservices Architecture to guarantee maximum fault tolerance and independent vertical scaling. If the hospital experiences an unexpected surge in communication traffic due to an emergency crisis, the chat service spins up additional instances autonomously without impacting the foundational database engines processing incoming patient check-ins.

- API Gateway Layer: A centralized Nginx reverse proxy serves as the singular entry point for all external client-side requests, handling incoming SSL/TLS termination, rate limiting to mitigate Distributed Denial of Service (DDoS) threats, and dynamic load balancing across active application containers.

- Core HMS Microservice: Built using an asynchronous runtime environment, this service processes all stateless RESTful requests associated with patient registration, doctor shift allocations, electronic prescriptions, and transactional billing workflows.
- Real-Time Communication Microservice: An isolated node instance dedicated solely to maintaining stateful, continuous Web Sockets connections. It shifts message payloads across instances via a Redis Pub/Sub backplane, preventing cross-thread blocking.

2.2 OpenCV Edge-Processing Pipeline

To eliminate the massive computational overhead and astronomical bandwidth costs associated with streaming raw, high-definition multi-channel video directly to cloud servers, Health-X implements a localized edge-computing pipeline for computer vision processing.



1. Ingestion: The system hooks into standard hospital IP closed-circuit cameras using the Real-Time Streaming Protocol (RTSP). Streams are captured at native frame rates and ingested frame-by-frame into memory blocks.
2. Downsampling and Grayscale Transformation: To maintain low execution latency, frames are downsampled from high-definition footprints down to uniform 640×480 configurations. The frame is then stripped of chroma data, yielding a single-channel 8-bit grayscale matrix ($Y = 0.299R + 0.587G + 0.114B$) to accelerate geometric pixel processing.
3. Temporal Noise Reduction: A localized Gaussian Filter kernel (21×21) is systematically run over the matrix. This eliminates ambient sensor noise, rapidly changing environmental shadows, and sudden shifts in room lighting conditions that could yield false-positive motion triggers.
4. Background Subtraction & Contours: The pipeline passes the cleaned matrix through a Mixture of Gaussians (MOG2) segmentation model to isolate the foreground moving

silhouette from static ward furniture. If a contour breaks a specific spatial threshold, an active bounding tracking array is drawn around the pixel matrix.

5. Anatomical Axis Evaluation: The geometric bounding box is evaluated frame-over-frame. When a patient sits up, stands, or falls, the system tracks the sudden transformation of the box's spatial coordinates and aspect ratio. A persistent delta exceeding safe baseline parameters triggers an instant localized IPC (Inter-Process Communication) event.

2.3 Real-Time Cryptographic Chat Protocol

The structural communication framework operates over persistent, bi-directional WebSockets (WSS) protocols, moving completely away from traditional, highly inefficient HTTP polling mechanics. To ensure absolute data confidentiality during high-stakes consultation routing:

- Handshake Protocol: When an authenticated staff member logs into the client-side system, a stateful connection request upgrades the HTTP connection to a secure WebSocket connection.
- Client-Side Cryptography: Message payloads containing sensitive patient telemetry or diagnostic charts are encrypted on the local device before serialization. The system uses a standard AES-256-GCM (Advanced Encryption Standard in Galois/Counter Mode) envelope, pairing it with an ephemeral initialization vector (IV) for each payload.
- Perfect Forward Secrecy (PFS): Keys are managed utilizing a custom variant of the Double Ratchet Algorithm. Even if an adversary intercepts a long-term cryptographic master key from a server image, past clinical conversations remain unreadable and perfectly secure.

3. Tools and Technologies

Health-X uses an enterprise-grade technology stack chosen specifically to minimize computational latency, maintain data integrity under heavy loads, and provide high cross-platform usability.

Category	Technology	Technical Rationale & Exact Implementations
Frontend UI Engine	React.js v18+ & Tailwind CSS	Utilizes Virtual DOM reconciliation and concurrent rendering modes to build high-performance dashboards that refresh clinical charts dynamically without UI freezing or stuttering.
State Management	Redux Toolkit	Serves as the central immutable client store, caching transient application states, managing active doctor shifts, and queuing incoming live network notifications.
Backend Architecture	Node.js (Express) & Python	Node.js: Drives the highly concurrent asynchronous I/O loops for admin management. FastAPI: Handles computational ML logic and heavy numerical video frame

Category	Technology	Technical Rationale & Exact Implementations
	(FastAPI)	indexing via native multi-threading layers.
Computer Vision Core	OpenCV, NumPy, MediaPipe	OpenCV: Manages baseline multi-channel RTSP stream decoding, frame operations, and image arithmetic. NumPy: Processes raw pixel structures as high-speed vector arrays. MediaPipe: Extracts skeletal tracking arrays for posture validation.
Real-Time Gateway	Socket.io / WebSockets	Provides a structured abstraction layer over baseline TCP sockets, adding automatic reconnection routines, heartbeat packets, and logical channel multiplexing.
Primary Persistence	PostgreSQL v15+	An ACID-compliant relational database engine enforcing strict foreign key constraints, complex indexing schemas, and binary JSON structures ($JSONB$) for flexible patient health charts.

4.Implementation

4.1 Core Hospital Management Engine

The core engine handles critical workflows by translating complex real-world hospital procedures into deterministic database transactions.

- Priority Queuing Algorithm: Patients are assigned an automated triage state based on vital data signs entered during reception registration. The scheduling service evaluates these variables and continuously re-orders incoming queues using an optimized min-heap priority framework, ensuring that life-critical anomalies bypass standard appointment backlogs.

- Automated Billing Engine: The ledger system ties an open transaction session to a patient's UUID upon admission. As the patient interacts with the ecosystem, background triggers log expenses: laboratory API results auto-inject itemized lab costs, pharmacy dispensaries log medication codes, and room management models register precise timestamps for bed occupancy. Upon discharge, a composite aggregation query computes the complete transactional summary, calculating real-time insurance deductible splits based on activ**e** pre-configured provider schemas.

4.2 Secure Intra-Hospital Communication

The communication module functions via real-time message routing brokers. When an OpenCV tracking node dispatches an alert payload to the API gateway, the gateway passes the data packet straight to the WebSocket microservice.

The service inspects the target ward ID against the active shift-roster database table in Redis. It instantly identifies the exact mobile and web socket connections mapped to the assigned duty doctors and nurses.

The payload bypasses general client chat message queues and displays on the recipient's device as a high-priority, persistent overlay notification accompanied by an audible alarm. This loop operates completely independently of external networks, maintaining continuity even during internet blackouts.

5. Major Findings / Outcomes / Results

5.1 System Performance Under Enterprise Load Simulation

To evaluate system stability under enterprise constraints, a load-testing harness simulated a highly active facility operating with 500 beds, 150 simultaneous operational system terminals, and 40 persistent OpenCV video input feeds.

EHR Query Execution Latency: With database tables scaled to hold over 500,000 legacy records, complex relational lookups achieved average response times of 34ms through appropriate multi-column indexing and materialized views.

- WebSocket Infrastructure Performance: The Redis-backed Socket.io server successfully distributed 8,000 concurrent messages per second during extreme spike simulations, maintaining a sub-15ms message routing delay with zero packet drop.

5.2 Computer Vision Accuracy and Validation Metrics

The OpenCV motion and anomaly detection module was rigorously benchmarked across a dataset consisting of 1,200 separate event actions simulated within a test ward environment across varied The results show that filtering through a dynamic temporal buffer (evaluating a minimum sequence of 15 frames) effectively reduces false positives caused by benign events—such as sudden medical blanket shifts or rapid movements by visiting family members—without delaying structural alert broadcasts.

6. Conclusion and Future Scope

6.1 Conclusion

The Health-X framework successfully demonstrates that modern web standards, secure real-time messaging, and localized computer vision can be combined into a robust, integrated hospital management ecosystem. Moving vision processing out of resource-heavy cloud models and onto the local facility edge completely eliminates recurring external data bandwidth costs. More importantly, it keeps critical patient

monitoring functional even during external network outages.

By unifying administrative databases with active, real-time alert mechanics, Health-X shifts the core role of a hospital info system from a passive historical ledger to an active, protective layer that directly reduces human error and shortens clinical response times.

6.2 Future Scope

The developmental trajectory of Health-X includes several key technological upgrades:

- Clinical NLP Transcription Engines: Integrating compact, locally hosted Large Language Models (LLMs) to listen to verified doctor-patient audio streams and automatically generate structured electronic clinical summaries, saving hours of manual data entry.

- Non-Contact Vital Sign Estimation : Updating the OpenCV processing pipelines to incorporate remote Photoplethysmography frameworks. This allows standard high-definition camera arrays to detect micro-variations in skin vascular reflectivity, providing real-time pulse and respiration tracking without requiring physical, wired sensors.

- FHIR Structural Standard Transition: Migrating all core data models to full HL7 FHIR (Fast Healthcare Interoperability Resources) compliance to enable native, seamless data exchange with international healthcare networks and public health repositories.

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7. References

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2. **Node.js Documentation – https://nodejs.org**
3. **React Documentation – https://react.dev**
4. **OpenJDK Documentation – https://openjdk.org**
5. **GNU g++ Compiler Documentation**

8. Appendices

- **Sample Code Snippets (C++/Java)**
- **API Endpoints Used**
- **Screenshots of Application Interface**
- **Docker Configuration Files**